Session 2: Multiple categorical variables.

Pre course: Statistics.

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Agenda

- Venn diagrams (e1).
- Contingency tables (e2).
- Visualizations for two categorical variables (e2).
- Exercises.

Basic idea for this session:

From analysis with only one categorical variable

→ to analysis with more than one categorical variables.

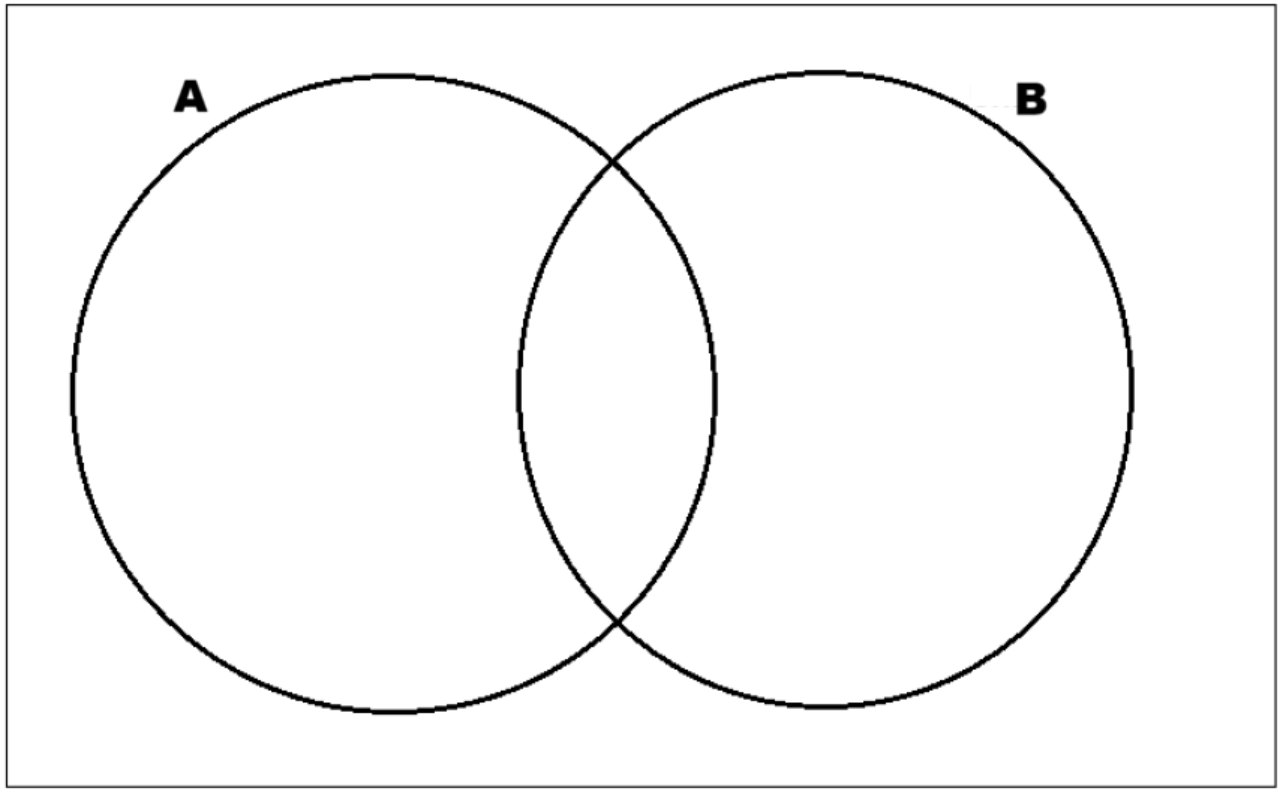
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Venn diagrams.

Notation: union (∪), intersection (∩), negation (’).

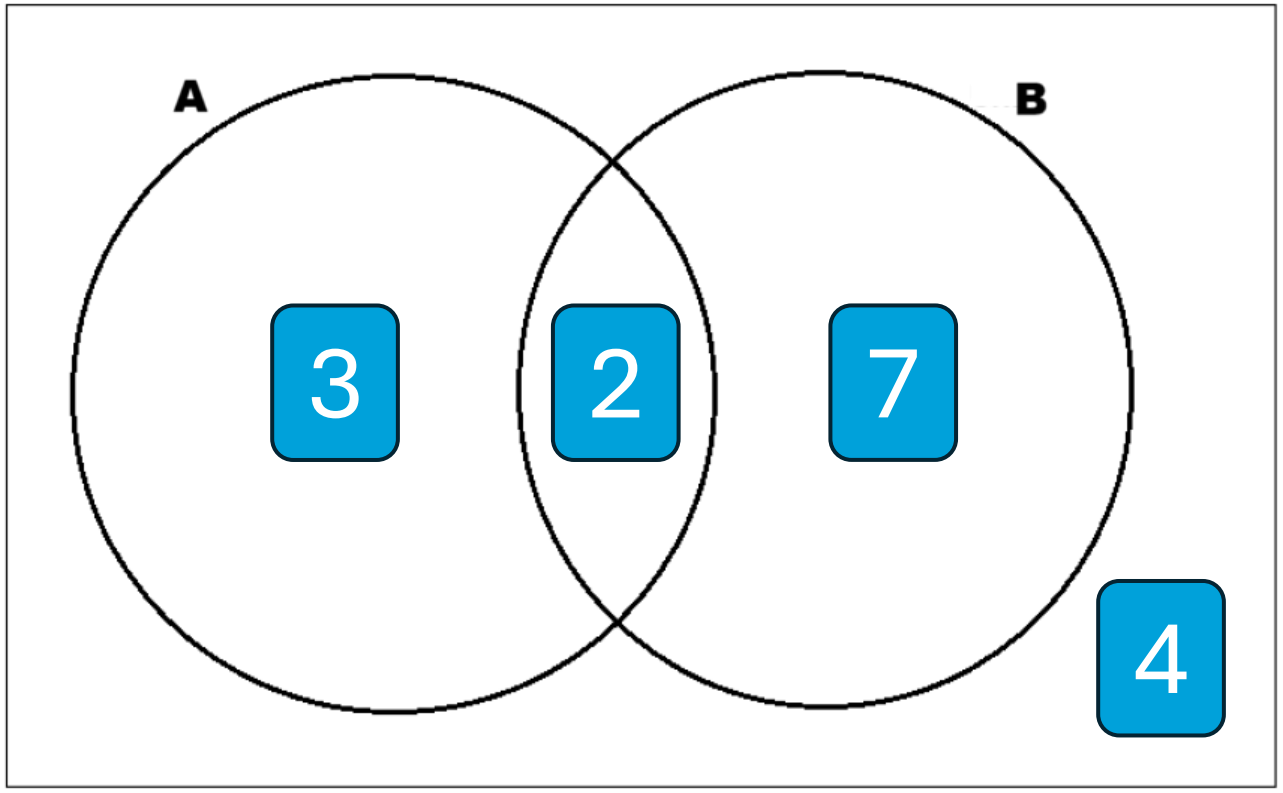
- (A) =
- (A ∪ B) =
- (A ∩ B) =
- (A)’ =
- (A ∪ B)’ =
- (A ∩ B)’ =
- U =
- U’ =



Venn diagrams.

Notation: union (∪), intersection (∩), negation (’).

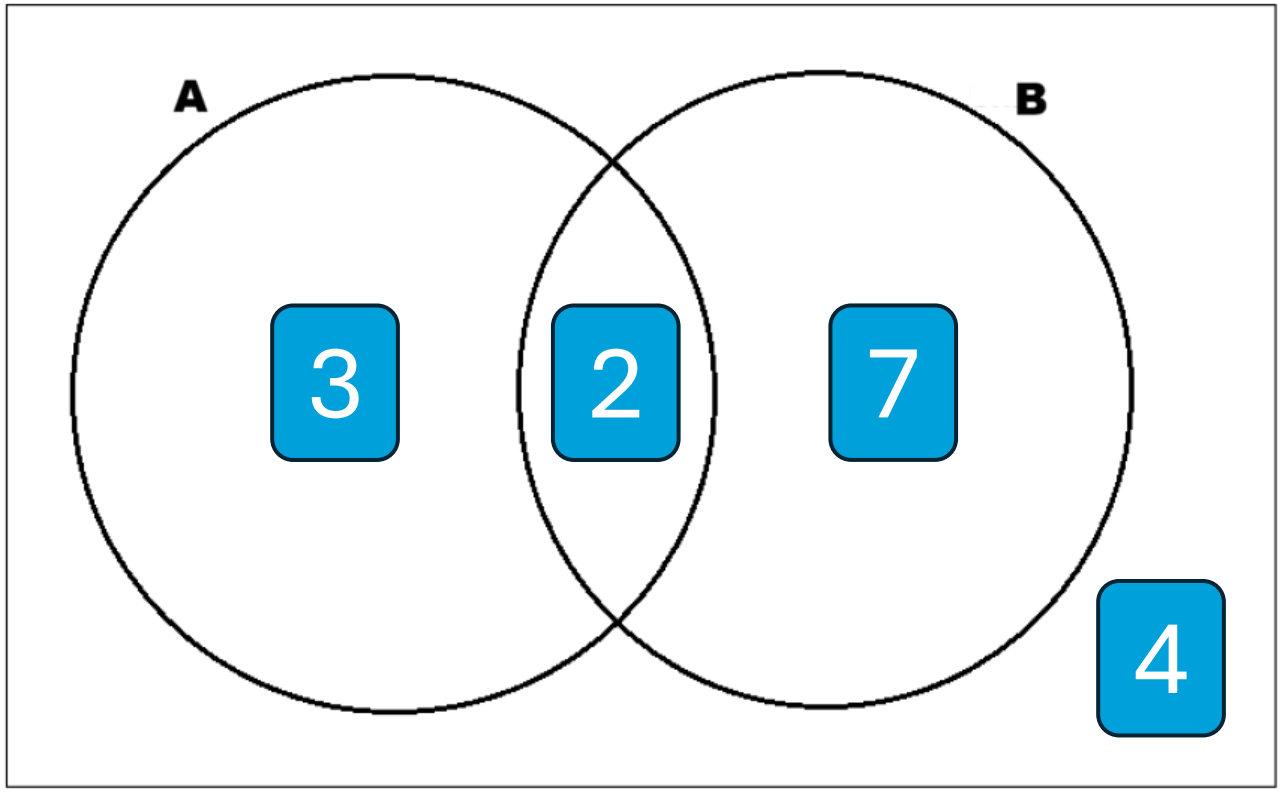
- (A) =
- (A ∪ B) =
- (A ∩ B) =
- (A)’ =
- (A ∪ B)’ =
- (A ∩ B)’ =
- U =
- U’ =



Venn diagrams.

Notation: union (∪), intersection (∩), negation (’).

- (A) = *3 + 2*
- (A ∪ B) = *3 + 2 + 7*
- (A ∩ B) = *2*
- (A)’ = *2 + 7 + 4*
- (A ∪ B)’ = *4*
- (A ∩ B)’ = *3 + 7 + 4*
- U = *3 + 2 + 7 + 4*
- U’ = ∅



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- Venn diagrams (e1).
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Contingency tables.

- Two way frequency table.
- Can be constructed with
  - Frequencies (N) or
  - Percentages.
- Marginal distribution (at the margins).
- Conditional distribution (conditional to some value)

Gender	Performs	Frequency
Female	Yes	35
Female	No	5
Male	Yes	25
Male	No	75

N		Perform care duties	
		Yes	No
Gender	Female	35	5
	Male	25	75

Contingency tables.

- Two way frequency table.
- Can be constructed with
  - Frequencies (N) or
  - Percentages.
- Marginal distribution (at the margins).
- Conditional distribution (conditional to some value)

Gender	Performs	Frequency
Female	Yes	35
Female	No	5
Male	Yes	25
Male	No	75

N		Perform care duties		Totals (by gender)
		Yes	No	
Gender	Female	35	5	40
	Male	25	75	100
Totals (by care duties)		60	80	140

Contingency tables.

- From N to Percentage of the total.

N		Perform care duties		Totals (by gender)
		Yes	No	
Gender	Female	35	5	**40**
	Male	25	75	**100**
Totals (by care duties)		**60**	**80**	**140**

%		Perform care duties		Percentage (by gender)
		Yes	No	
Gender	Female	0,25	0,04	**29**
	Male	0,18	0,54	**71**
Percentage (by care duties)		**42,9**	**57,1**	**100**

Contingency tables.

- Marginal distribution (at the margin).

N		Perform care duties		Totals (by gender)
		Yes	No	
Gender	Female	35	5	**40**
	Male	25	75	**100**
Totals (by care duties)		**60**	**80**	**140**

%		Perform care duties		Totals
		Yes	No	
Gender	Female	87,5	12,5	**100**
	Male	25,0	75,0	**100**
Percentage (by care duties)		**42,9**	**57,1**	**100**

%		Perform care duties		Percentage (by gender)
		Yes	No	
Gender	Female	58,3	6,3	**28,6**
	Male	41,7	93,8	**71,4**
Totals		**100**	**100**	**100**

Contingency tables.

- Conditional distribution (conditional to some value).
- For example, conditional to:
  - Being a man, or
  - Not performing care duties.

N		Perform care duties		Totals (by gender)
		Yes	No	
Gender	Female	35	5	40
	Male	25	75	100
Totals (by care duties)		60	80	140

%		Perform care duties		Totals
		Yes	No	
Gender	Female	87,5	12,5	100
	Male	25,0	75,0	100
Percentage (by care duties)		42,9	57,1	100

%		Perform care duties		Percentage (by gender)
		Yes	No	
Gender	Female	58,3	6,3	28,6
	Male	41,7	93,8	71,4
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- **Visualizations for two categorical variables (e2).**
- Exercises.

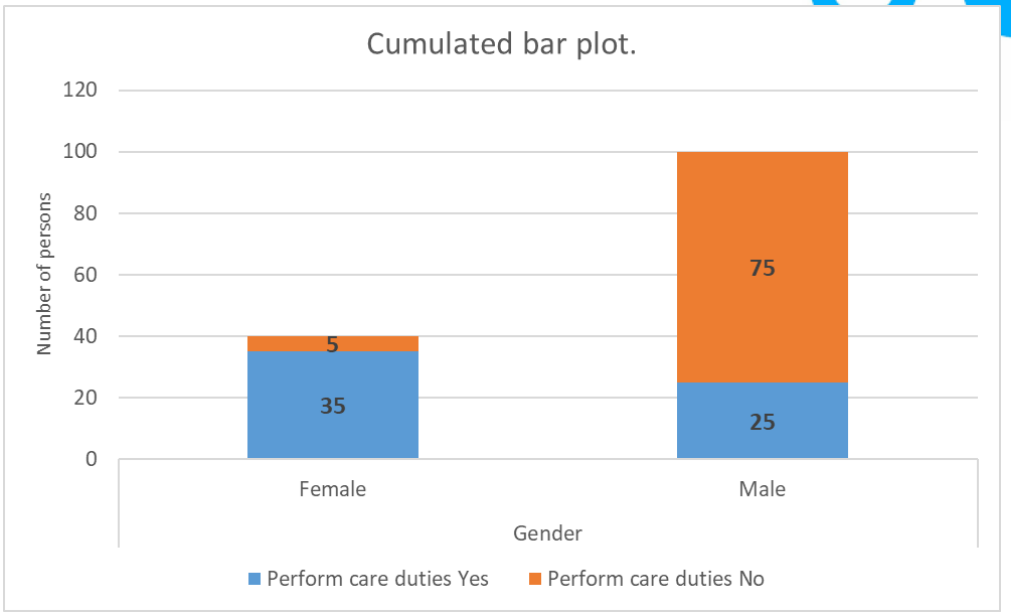
Visualizations.

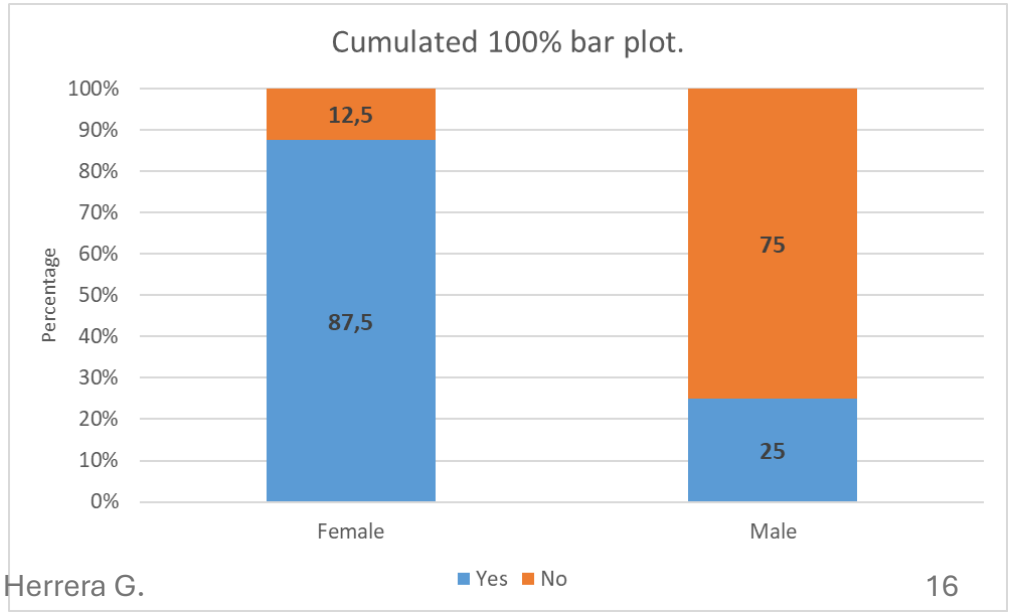
- Cumulated bar charts.
- Cumulated 100% bar charts.
- Mosaic plots.

Visualizations.

N		Perform care duties		Totals (by gender)
		Yes	No	
Gender	Female	35	5	40
	Male	25	75	100
Totals (by care duties)		60	80	140

%		Perform care duties		Totals
		Yes	No	
Gender	Female	87,5	12,5	100
	Male	25,0	75,0	100
Percentage (by care duties)		42,9	57,1	100





Visualizations.

Figure 1. Cumulated Bar Plot.

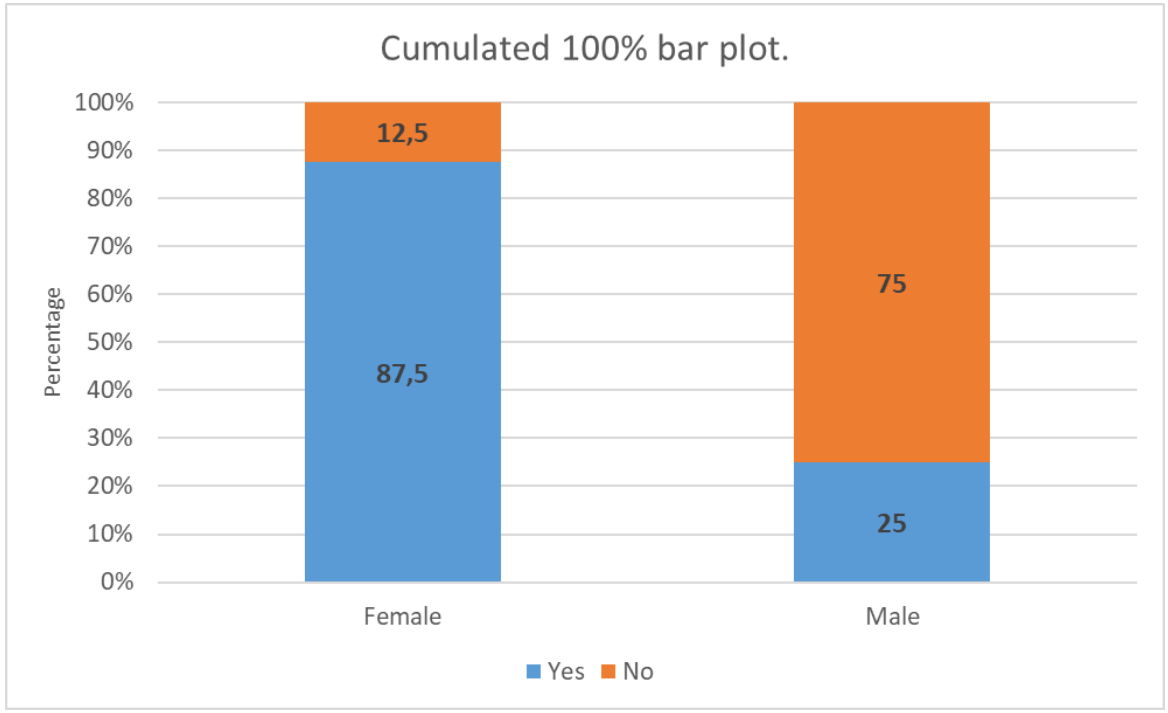
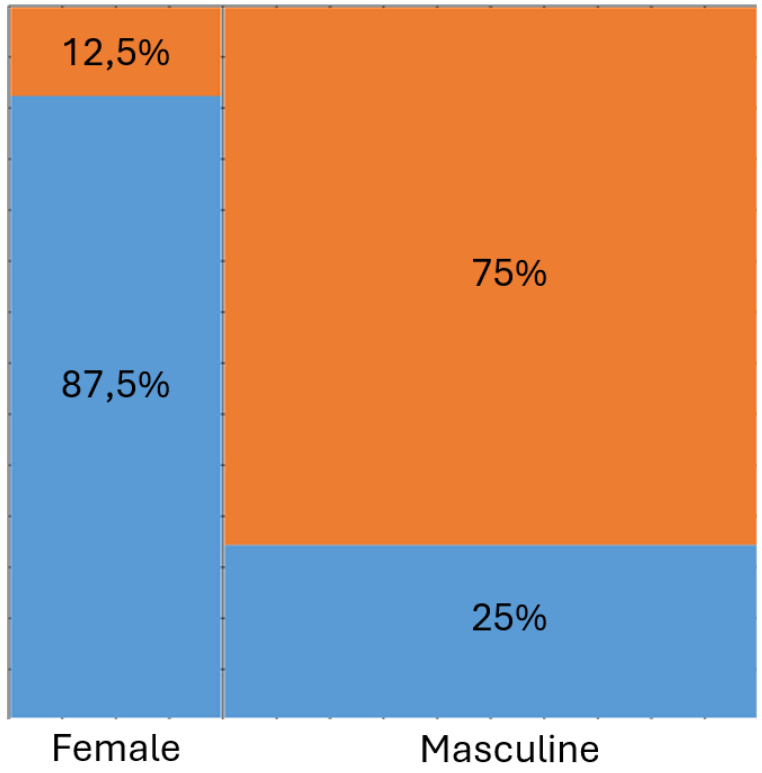


Figure 2. Mosaic Plot.



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- **Exercises.**

Exercises.

- We covered all the exercises ☺
- Any questions?

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